Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Gabriel Machado & Sons Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
12464 S Vincent RD	Turlock	Merced	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1970

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X041-X130-X008-XXXX

B. OPERATORS

Machado, David

Operator name: Machado, David

Telephone no.: (209) 632-3068 Landline, (209) 678-2515 Cellular

Mailing Address Number and Street	City	State	Zip Code
12464 Vincent RD	Turlock	CA	95380

Machado, Gabe

Operator name: Machado, Gabe

Telephone no.: (209) 634-2078 Landline, (209) 678-2531 Cellular

Mailing Address Number and Street	City	State	Zip Code
12464 Vincent RD	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

Machado, David

Legal owner name: Machado, David

Telephone no.: (209) 632-3068 Landline, (209) 678-2515 Cellular

Mailing Address Number and Street	City	State	Zip Code
12464 Vincent RD	Turlock	CA	95380

Machado, Gabe

Legal owner name:	Machado, Gabe	Telephone no.:	(209) 634-2078	(209) 678-2531
			Landline	Cellular
12464 Vincent RD	Turlock		CA	95380
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

Machado, Mary

Legal owner name:	Machado, Mary	Telephone no.:	(209) 678-2531	
			Landline	Cellular
12464 Vincent RD	Turlock		CA	95380
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	75	204	170	0	0
Number under roof	597	0	0	0	135	130
Maximum number	597	75	204	170	135	130
Average number	597	75	204	170	135	130
Avg live weight (lbs)	1,300	1,400	975	750		

Predominant milk cow breed: Holstein

Average milk production: 94 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 22,758.64 *tons per reporting period*

Total nitrogen from manure: 284,929.26 *lbs per reporting period*

After ammonia losses (30% loss applied): 199,450.48 *lbs per reporting period*

Total phosphorus from manure: 46,709.71 *lbs per reporting period*

Total potassium from manure: 123,744.50 *lbs per reporting period*

Total salt from manure: 298,343.70 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 10,140,000 *gallons*

Total nitrogen generated: 49,335.57 *lbs*

Total phosphorus generated: 6,195.26 *lbs*

Total potassium generated: 68,497.76 *lbs*

Total salt generated: 342,398.69 *lbs*

	10,140,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	10,140,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
DW - Well #1 (West)	Ground water
DW - Well #2 (East)	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	Corral solids	2,000.00 *ton*	Dry-weight	13.0		25,300.00	7,900.00	20,000.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	88,044.00	27,492.00	69,600.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	88,044.00	27,492.00	69,600.00	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #1	40	40	2	process wastewater	X041-X130-X008-XXXX
Field #2	20	20	2	process wastewater	X041-X130-X008-XXXX
Field #3	30	30	2	process wastewater	X041-X130-X075-XXXX
Totals for areas that were used for application	90	90	6		
Totals for areas that were not used for application					
Land application area totals	90	90	6		

B. CROPS AND HARVESTS

Field #1

Field name: Field #1

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	480.00 *ton*	Dry-weight		70.3	28,900.00	4,400.00	38,500.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.00	206.00	31.36	274.43	798.34

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,040.00 *ton*	Dry-weight		56.1	13,400.00	2,700.00	14,900.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.00	305.90	61.64	340.14	1,187.06

Field #2

Field name: Field #2

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	205.00 *ton*	Dry-weight		69.4	32,600.00	5,100.00	42,500.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.25	204.50	31.99	266.60	777.85

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	530.00 *ton*	Dry-weight		58.7	11,400.00	2,200.00	15,200.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.50	249.53	48.16	332.71	1,138.23

Field #3

Field name: Field #3

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	321.00 *ton*	Dry-weight		70.4	29,800.00	4,400.00	42,100.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.70	188.77	27.87	266.68	753.79

Field #3

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	838.00 *ton*	Dry-weight		61.4	12,300.00	2,300.00	15,300.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.93	265.24	49.60	329.94	1,142.92

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #1 - 11/15/2022: Oats, silage-soft dough

Field name: Field #1

Crop: Oats, silage-soft dough — Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.88	14.33	162.13	830.29	990,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.24	5,758,425.00 *gal*
Application event totals		107.88	14.33	162.13	861.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	114.41	15.20	171.96	880.61	1,050,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.24	5,758,425.00 *gal*
Application event totals		114.41	15.20	171.96	911.84	

Field #1 - 06/24/2023: Corn, silage

Field name: Field #1

Crop: Corn, silage — Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	127.22	15.59	153.01	700.60	870,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	32.33	5,960,475.00 *gal*
Application event totals		127.22	15.59	153.01	732.94	

Field #1 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	122.83	15.05	147.73	676.45	840,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	26.30	4,849,200.00 *gal*
Application event totals		122.83	15.05	147.73	702.75	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.52	6,364,575.00 *gal*
Application event totals		0.00	0.00	0.00	34.52	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.52	6,364,575.00 *gal*
Application event totals		0.00	0.00	0.00	34.52	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.43	6,162,525.00 *gal*
Application event totals		0.00	0.00	0.00	33.43	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.88	14.33	162.13	830.29	990,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.71	6,768,675.00 *gal*
Application event totals		107.88	14.33	162.13	867.00	

09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.78	5,859,450.00 *gal*
Application event totals		0.00	0.00	0.00	31.78	

Field #2 - 11/15/2022: Oats, silage-soft dough

Field name: Field #2

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.99	11.29	127.74	654.16	390,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	29.59	2,727,675.00 *gal*
Application event totals		84.99	11.29	127.74	683.76	

03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.07	13.03	147.39	754.81	450,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.78	2,929,725.00 *gal*
Application event totals		98.07	13.03	147.39	786.59	

Field #2 - 06/24/2023: Corn, silage

Field name: Field #2

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	122.83	15.05	147.73	676.45	420,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.98	3,131,775.00 *gal*
Application event totals		122.83	15.05	147.73	710.42	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	32.88	3,030,750.00 *gal*
Application event totals		0.00	0.00	0.00	32.88	

Field #2 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.26	3,434,850.00 *gal*
Application event totals		0.00	0.00	0.00	37.26	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.07	13.03	147.39	754.81	450,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.07	3,232,800.00 *gal*
Application event totals		98.07	13.03	147.39	789.88	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	33.98	3,131,775.00 *gal*
Application event totals		0.00	0.00	0.00	33.98	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.53	12.16	137.57	704.48	420,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.26	3,434,850.00 *gal*
Application event totals		91.53	12.16	137.57	741.75	

09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.36	3,535,875.00 *gal*
Application event totals		0.00	0.00	0.00	38.36	

Field #3 - 11/15/2022: Oats, silage-soft dough

Field name: Field #3

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Field #3 - 11/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.53	12.16	137.57	704.48	630,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.65	5,758,425.00 *gal*
Application event totals		91.53	12.16	137.57	746.13	

03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.96	14.48	163.77	838.67	750,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.03	6,364,575.00 *gal*
Application event totals		108.96	14.48	163.77	884.70	

Field #3 - 06/24/2023: Corn, silage

Field name: Field #3

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.34	4,748,175.00 *gal*
Application event totals		0.00	0.00	0.00	34.34	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	133.07	11.11	169.64	932.64	660,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	32.15	4,445,100.00 *gal*
Application event totals		133.07	11.11	169.64	964.79	

Field #3 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.42	4,344,075.00 *gal*
Application event totals		0.00	0.00	0.00	31.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	111.13	13.62	133.66	612.02	570,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	31.42	4,344,075.00 *gal*
Application event totals		111.13	13.62	133.66	643.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	32.88	4,546,125.00 *gal*
Application event totals		0.00	0.00	0.00	32.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.89	12.74	144.12	738.03	660,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	34.34	4,748,175.00 *gal*
Application event totals		95.89	12.74	144.12	772.37	

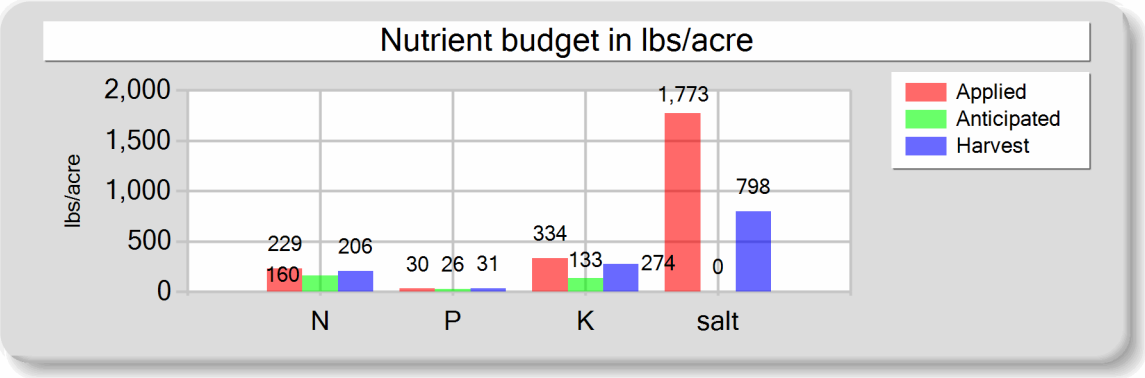
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	35.80	4,950,225.00 *gal*
Application event totals		0.00	0.00	0.00	35.80	

B. NUTRIENT BUDGET

Field #1 - 11/15/2022: Oats, silage-soft dough

Field name: Field #1 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	222.29	29.54	334.09	1,710.89
Fresh water	0.00	0.00	0.00	62.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.29	29.54	334.09	1,773.36
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	206.00	31.36	274.43	798.34
Nutrient balance	23.29	-1.83	59.66	975.03
Applied to removed ratio	1.11	0.94	1.22	2.22

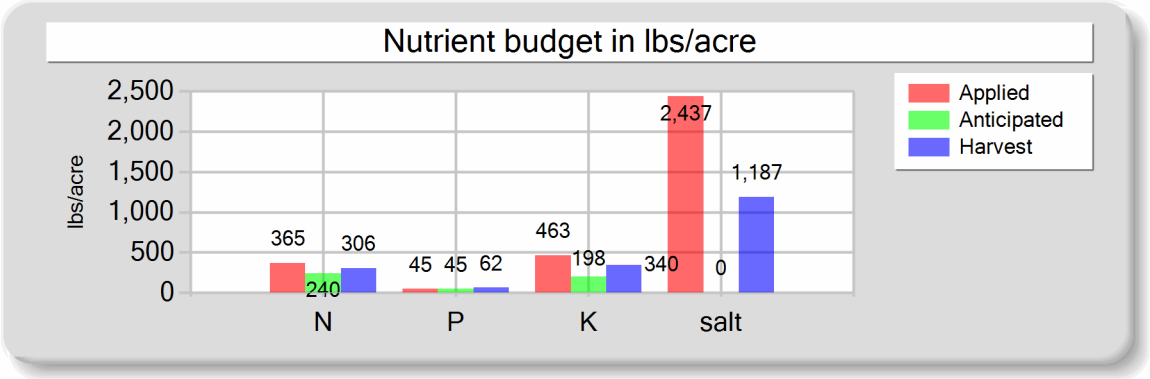
Fresh water applied	
11,516,850.00	*gallons*
424.13	*acre-inches*
10.60	*inches/acre*

Process wastewater applied	
2,040,000.00	*gallons*
75.13	*acre-inches*
1.88	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #1 - 06/24/2023: Corn, silage

Field name: Field #1 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	357.92	44.98	462.87	2,207.34
Fresh water	0.00	0.00	0.00	229.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	364.92	44.98	462.87	2,436.94
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	305.90	61.64	340.14	1,187.06
Nutrient balance	59.03	-16.66	122.73	1,249.89
Applied to removed ratio	1.19	0.73	1.36	2.05

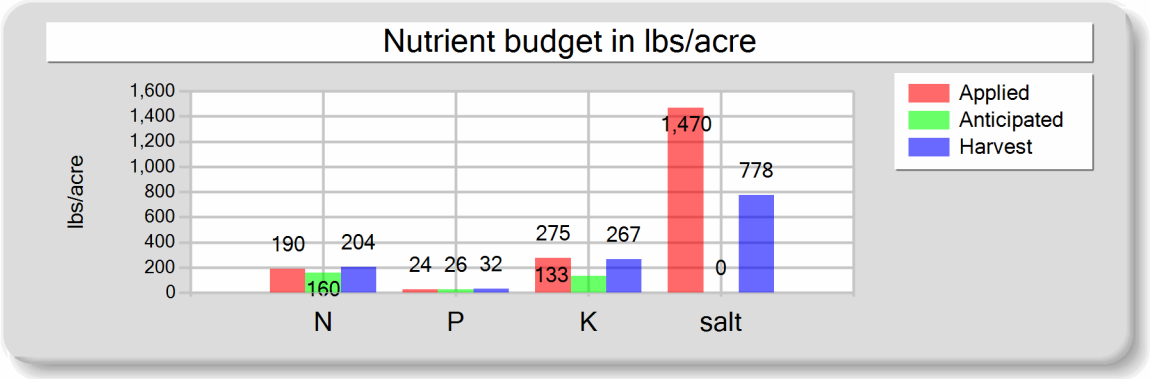
Fresh water applied	
42,329,475.00	*gallons*
1,558.85	*acre-inches*
38.97	*inches/acre*

Process wastewater applied	
2,700,000.00	*gallons*
99.43	*acre-inches*
2.49	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 11/15/2022: Oats, silage-soft dough

Field name: Field #2 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.06	24.32	275.13	1,408.97
Fresh water	0.00	0.00	0.00	61.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	190.06	24.32	275.13	1,470.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	204.50	31.99	266.60	777.85
Nutrient balance	-14.44	-7.67	8.53	692.49
Applied to removed ratio	0.93	0.76	1.03	1.89

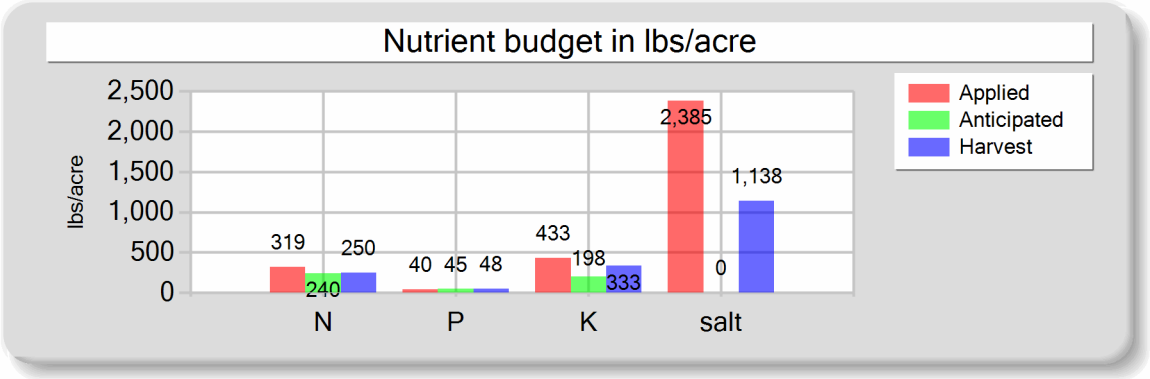
Fresh water applied	
5,657,400.00	*gallons*
208.34	*acre-inches*
10.42	*inches/acre*

Process wastewater applied	
840,000.00	*gallons*
30.93	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #2 - 06/24/2023: Corn, silage

Field name: Field #2 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	312.43	40.25	432.69	2,135.74
Fresh water	0.00	0.00	0.00	248.79
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	319.43	40.25	432.69	2,384.52
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.53	48.16	332.71	1,138.23
Nutrient balance	69.89	-7.91	99.98	1,246.29
Applied to removed ratio	1.28	0.84	1.30	2.09

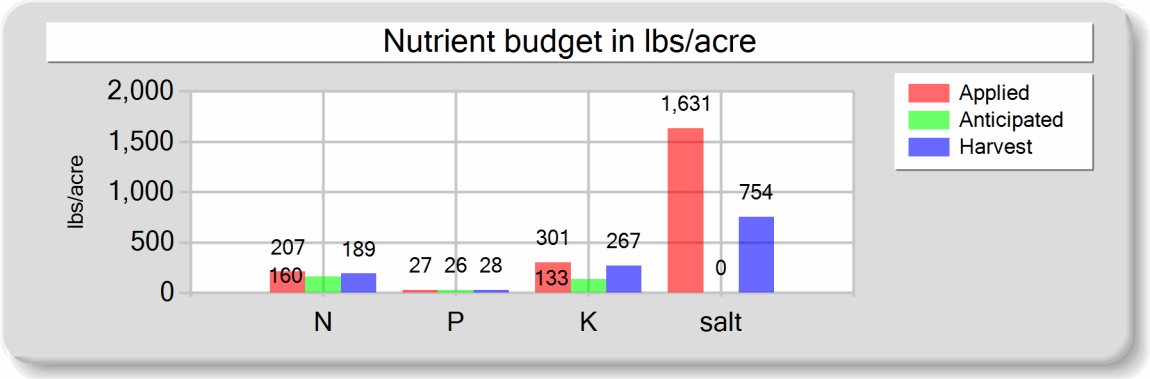
Fresh water applied	
22,932,675.00	*gallons*
844.53	*acre-inches*
42.23	*inches/acre*

Process wastewater applied	
1,290,000.00	*gallons*
47.51	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #3 - 11/15/2022: Oats, silage-soft dough

Field name: Field #3 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	200.50	26.64	301.34	1,543.16
Fresh water	0.00	0.00	0.00	87.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.50	26.64	301.34	1,630.83
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	188.77	27.87	266.68	753.79
Nutrient balance	18.73	-1.23	34.66	877.04
Applied to removed ratio	1.10	0.96	1.13	2.16

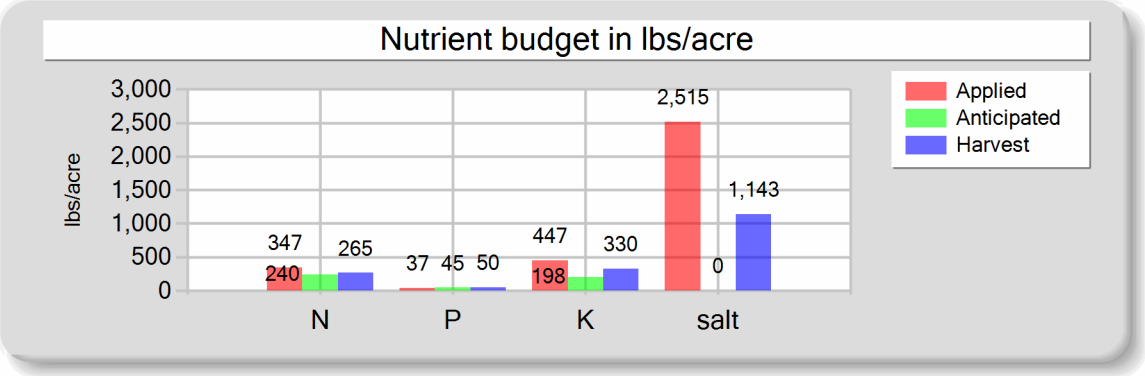
Fresh water applied	
12,123,000.00	*gallons*
446.45	*acre-inches*
14.88	*inches/acre*

Process wastewater applied	
1,380,000.00	*gallons*
50.82	*acre-inches*
1.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field #3 - 06/24/2023: Corn, silage

Field name: Field #3 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	340.09	37.47	447.42	2,282.69
Fresh water	0.00	0.00	0.00	232.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	347.09	37.47	447.42	2,515.04
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	265.24	49.60	329.94	1,142.92
Nutrient balance	81.84	-12.13	117.48	1,372.12
Applied to removed ratio	1.31	0.76	1.36	2.20

Fresh water applied	
32,125,950.00	*gallons*
1,183.09	*acre-inches*
39.44	*inches/acre*

Process wastewater applied	
1,890,000.00	*gallons*
69.60	*acre-inches*
2.32	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 19.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	7,900.00	26,800.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	7,900.00	20,000.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,570.00	466.00	0.00	1.80	87.60	838.00								9,100.00	3,200
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	700.00	469.00	0.00	0.90	85.90	843.00								9,160.00	3,860
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.90

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	521.00	443.00	0.00	1.30	69.40	785.00								7,810.00	4,020
DL	1.00	0.50	0.01	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	724.00	508.00	0.00	0.80	60.50	924.00								9,840.00	5,080
DL	1.00	0.50	0.01	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

DW - Well #1 (West)

DW - Well #1 (West)

Well #1 (West)

Sample description: Well #1 (West)

Sample date: 11/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			16.40								546.00	
DL			0.10								10.00	

DW - Well #2 (East)

Well #2 (East)

Sample description: Well #2 (East)

Sample date: 11/27/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value			16.20								542.00	
DL			0.10								10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/24/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										32.50	26
DL	1.00										10.00	10

D. SOIL ANALYSES

Field #1

Field 1

Sample and source description: Field 1

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			91.00				
DL			2.00				

Field #2

Field 2

Sample and source description: Field 2

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			104.00				
DL			2.00				

Field #3

Field 3

Sample and source description: Field 3

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			60.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Field #1 - 11/15/2022: Oats, silage-soft dough

Field #1 - 11/15/2022: Oats, silage-soft dough

Field #1

Sample and source description: Field #1

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,900.00	4,400.00	38,500.00		11.20
DL	100.00	100.00	100.00		0.01

Field #1 - 06/24/2023: Corn, silage

Field #1

Sample and source description: Field #1

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,700.00	14,900.00		5.20
DL	100.00	100.00	100.00		0.01

Field #2 - 11/15/2022: Oats, silage-soft dough

Field #2

Sample and source description: Field #2

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,600.00	5,100.00	42,500.00		12.40
DL	100.00	100.00	100.00		0.01

Field #2 - 06/24/2023: Corn, silage

Field #2

Sample and source description: Field #2

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,200.00	15,200.00		5.20
DL	100.00	100.00	100.00		0.01

Field #3 - 11/15/2022: Oats, silage-soft dough

Field #3

Sample and source description: Field #3

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,800.00	4,400.00	42,100.00		11.90
DL	100.00	100.00	100.00		0.01

Field #3 - 06/24/2023: Corn, silage

Field #3

Sample and source description: Field #3

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	2,300.00	15,300.00		5.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

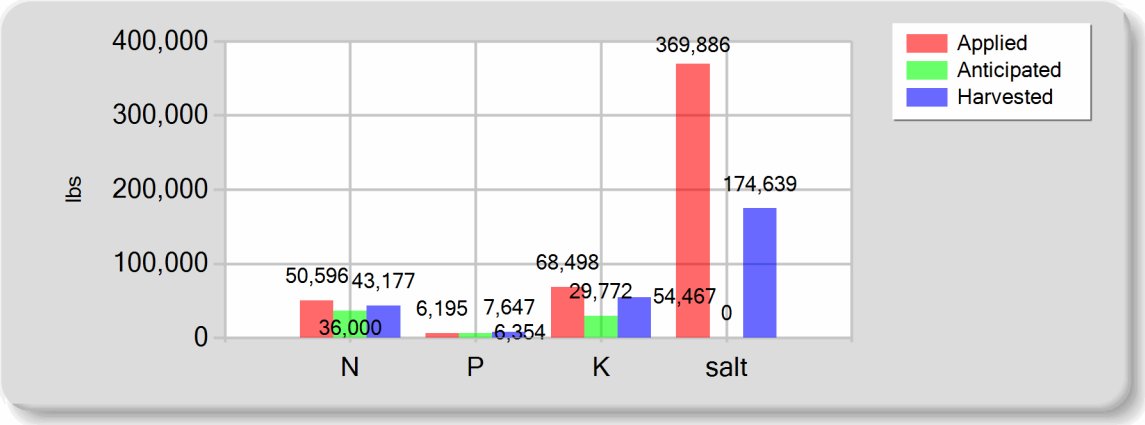
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

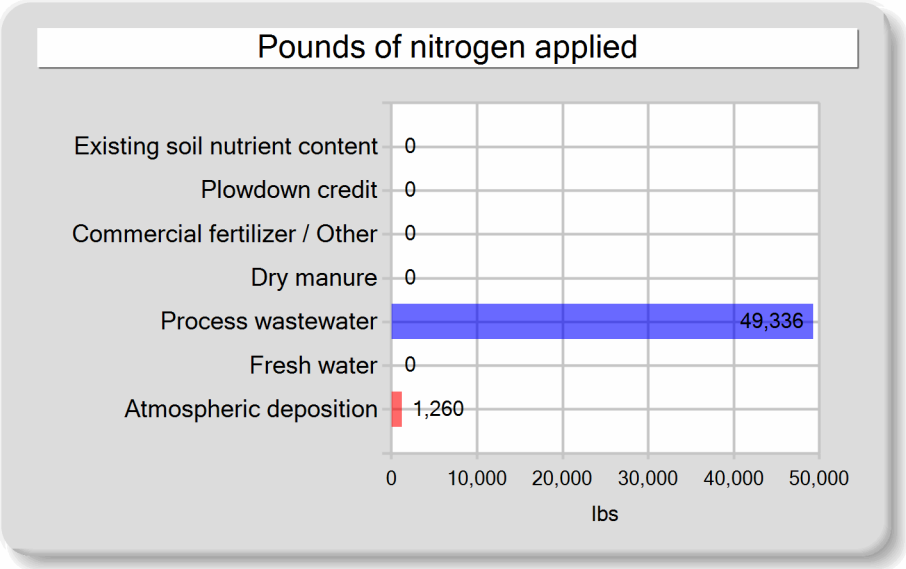
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

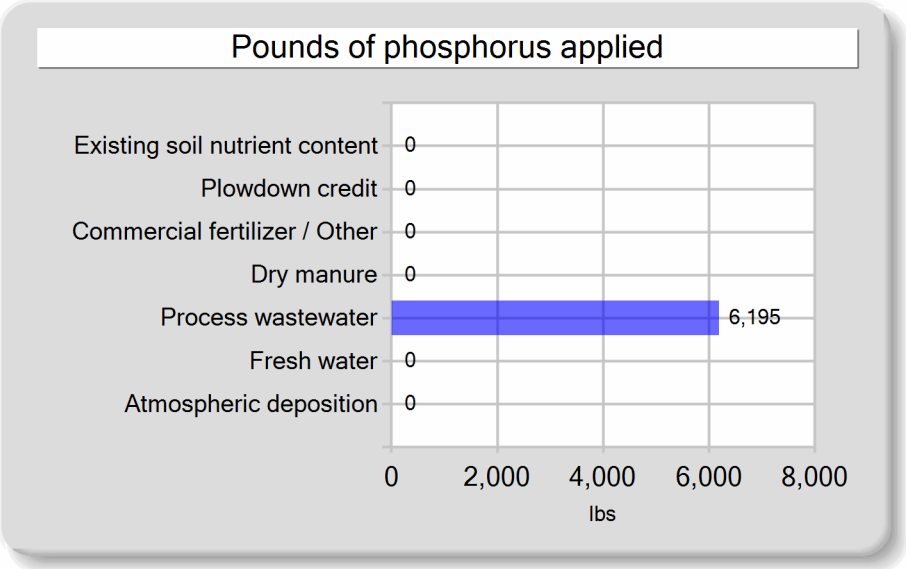
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	49,335.57	6,195.26	68,497.76	342,398.69
Fresh water	0.00	0.00	0.00	27,486.92
Atmospheric deposition	1,260.00	0.00	0.00	0.00
Total nutrients applied	50,595.57	6,195.26	68,497.76	369,885.61
Anticipated crop nutrient removal	36,000.00	6,354.00	29,772.00	0.00
Actual crop nutrient removal	43,176.73	7,647.01	54,467.38	174,638.70
Nutrient balance	7,418.84	-1,451.75	14,030.38	195,246.91
Applied to removed ratio	1.17	0.81	1.26	2.12

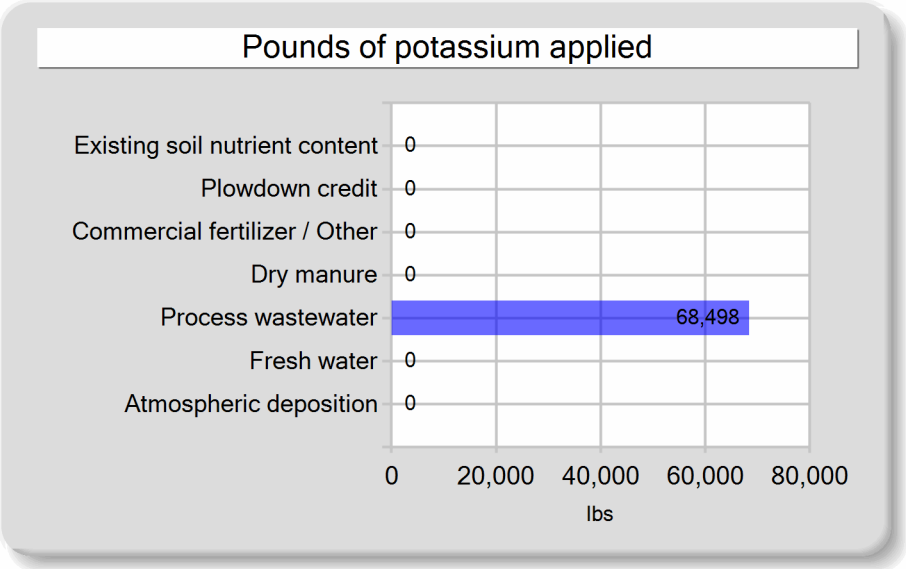
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

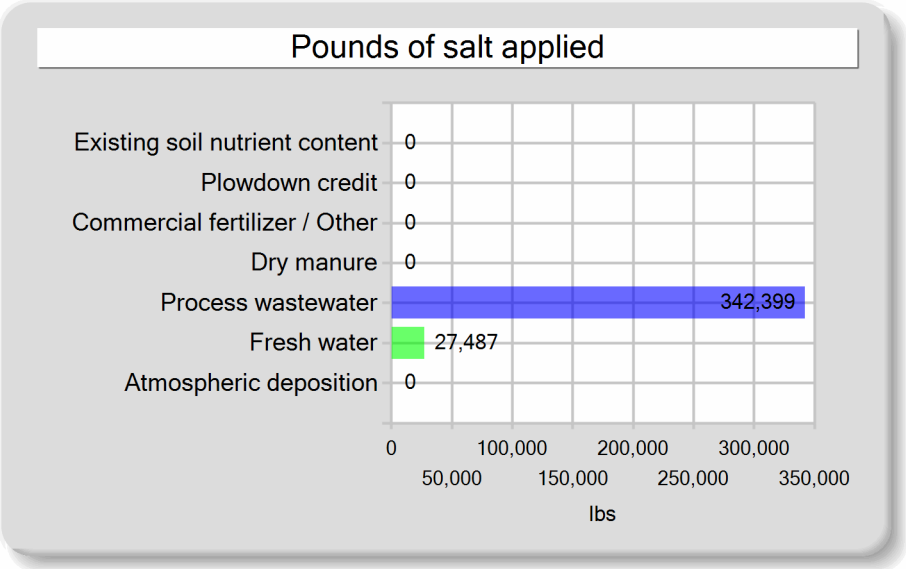


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Gabe Machado	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-21-24	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator:	**Gabe or Dave Machado**
Name of Dairy Facility:	**Gabe Machado & Sons Dairy**
Facility Address:	**12266 S. Vincent Rd, Turlock, CA 95380**
Contact Person Name and Phone Number:	**Gabe Machado (209) 678-2531**

Manure Hauler Information:

Name of Hauling Company/Person:	**Dores Ag Services, Inc.**
Address of Hauling Company /Person:	**22846 Second Avenue, Stevinson, CA 95374**
Contact Person and Phone Number:	**Phil Dores (209) 649-4503**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Silva Brothers	**224 S. Roselawn Ave**	**Turlock**	**95380**	**(209) 495-9669**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
224 S. Roselawn Ave	**Turlock**	**95380**	

Dates Hauled: **5/18/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **2,000** Tons

Manure Solids Content (if amount reported in tons): **87.0%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: _____ Date: 5/18/2023

Hauler's Signature: _____ Date: 5/18/2023

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Gabe Machado & Sons

Received: 11/28/2023 13:50
Reported: 12/06/2023 09:14

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K1223-01	Well #1 (West)	Ag Water	T. Madruga		11/27/2023 18:00
23K1223-02	Well #2 (East)	Ag Water	T. Madruga		11/27/2023 17:00

Default Cooler Temperature on Receipt °C: 1.7
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Sample Results

Sample: Well #1 (West)
23K1223-01 (Water)

Sampled: 11/27/2023 18:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.55**	mmhos/cm	0.01	1		11/30/23 15:09	SM 2510 B		BEK0875
Electrical Conductivity umhos	**546**	umhos/cm	10.0	1		11/30/23 15:09	SM 2510 B		BEK0875
Ammonia (as N)	ND	mg/L	0.00	1		11/27/23 18:00	Field		BEK0804
Nitrate Nitrogen as NO3N	**16.4**	mg/L	0.1	1	10	11/29/23 02:36	EPA 300.0		BEK0809
Temperature	**25.0**	units	0.0	1		11/30/23 15:09	SM 4500-H+	H	BEK0875
pH	**7.8**	units	1.0	1		11/30/23 15:09	SM 4500-H+	H	BEK0875

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Sample Results
(Continued)

Sample: Well #2 (East)
23K1223-02 (Water)

Sampled: 11/27/2023 17:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.54**	mmhos/cm	0.01	1		11/30/23 15:11	SM 2510 B		BEK0875
Electrical Conductivity umhos	**542**	umhos/cm	10.0	1		11/30/23 15:11	SM 2510 B		BEK0875
Ammonia (as N)	ND	mg/L	0.00	1		11/27/23 17:00	Field		BEK0804
Nitrate Nitrogen as NO3N	**16.2**	mg/L	0.1	1	10	11/29/23 02:57	EPA 300.0		BEK0809
Temperature	**25.0**	units	0.0	1		11/30/23 15:11	SM 4500-H+	H	BEK0875
pH	**7.6**	units	1.0	1		11/30/23 15:11	SM 4500-H+	H	BEK0875

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Hilmar, CA 95324

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Ranch: Gabe Machado & Sons

Received: 11/28/2023 13:50
Reported: 12/06/2023 09:14

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0809										
Blank (BEK0809-BLK1)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0809-BLK2)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0809-BLK3)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0809-BS1)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		103	90-110		
LCS (BEK0809-BS2)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	5.5		0.1	mg/L	5.000		110	90-110		
Duplicate (BEK0809-DUP1)		**Source: 23K1216-02**			Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	4.1		0.1	mg/L		4.0			1.74	10
Duplicate (BEK0809-DUP2)		**Source: 23K1221-02**			Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L		4.9			0.123	10
Matrix Spike (BEK0809-MS1)		**Source: 23K1216-02**			Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	9.2		0.1	mg/L	5.000	4.0	104	90-110		
Matrix Spike (BEK0809-MS2)		**Source: 23K1221-02**			Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	10.2		0.1	mg/L	5.000	4.9	107	90-110		
Reference (BEK0809-SRM1)					Prepared & Analyzed: 11/28/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0809-SRM2)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0809-SRM3)					Prepared: 11/28/2023 Analyzed: 11/29/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Ranch: Gabe Machado & Sons

Received: 11/28/2023 13:50
Reported: 12/06/2023 09:14

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0875										
Blank (BEK0875-BLK1)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.4		1.0	units						
Blank (BEK0875-BLK2)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.9		1.0	units						
Blank (BEK0875-BLK3)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.5		1.0	units						
Duplicate (BEK0875-DUP1)		**Source: 23K1231-01**			Prepared: 11/29/2023	Analyzed: 11/30/2023				
Electrical Conductivity	12.1		0.01	mmhos/cm		12.0			0.828	10
pH	7.4		1.0	units		7.4			0.00	10
Electrical Conductivity umhos	12100		10.0	umhos/cm		12000			0.829	10
Duplicate (BEK0875-DUP2)		**Source: 23K1350-01**			Prepared: 11/29/2023	Analyzed: 11/30/2023				
Electrical Conductivity	0.67		0.01	mmhos/cm		0.68			1.09	10
Electrical Conductivity umhos	672		10.0	umhos/cm		680			1.09	10
pH	6.3		1.0	units		6.3			0.159	10
Reference (BEK0875-SRM1)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
Electrical Conductivity	435			umhos/cm	426.0		102	90-110		
Reference (BEK0875-SRM2)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0875-SRM3)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEK0875-SRM4)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0875-SRM5)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0875 (Continued)										
Reference (BEK0875-SRM5)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEK0875-SRM6)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0875-SRM7)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0875-SRM8)					Prepared: 11/29/2023	Analyzed: 11/30/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/28/23 13:50 23K1223

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples 2 No. of Bottles: 2

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com
COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Gabe Machado & Sons
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:
Analysis and Bottles Required: (Please indicate Analysis)
(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Well #1 (West)	11/27/2023	1800	1.7	ND
2	Well #2 (East)	11/27/2023	1700	1.7	ND
3					
4					
5					
6					
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/27/2023 @ 1800	11/28/2023 @ 900
Second	JCP	Dellavalle Laboratory, Inc.	11/28/2023 @ 900	11-28-23 1:20
Third				
Fourth		DLI	11/28/23 13:50	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding a[illegible] under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all [illegible] of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

Invoicing Information:		*Shipping*	
Sampling hrs		$	In
Miles		$	Out
Consulting			
Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/28/23 13:50 23K1223

Shipping Information: Fed Ex □ UPS □ GSO □ Picked-Up ☒ Walk In □ DLI Sampler □
Ontrac □ PMS □ USPS □ Other □ _____

Shipping Container: Ice Chest ☒ None □ Box □

Refrigerant: Wet Ice: ☒ Blue Ice: □ None: □

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										